

Project Design Phase

Solution Architecture

Date	03 October 2026
Team ID	SWTID-2026-4228
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture bridges the employee-data management problem with a structured ServiceNow technology workflow. It uses an intermediate Import Set staging layer and a Transform Map processing layer to move spreadsheet data safely into the target Employee Test table. The architecture defines the data flow, field mapping, deduplication behavior, and controlled record creation or update process.

Its goals are to:

- Automate and streamline bulk ingestion of employee records from external Excel/CSV spreadsheets.
- Describe the complete data flow from raw spreadsheet upload through staging, transformation, and target-table population.
- Define field mapping and Coalesce rules required for accurate transformation and duplicate prevention.
- Provide a controlled and traceable process for importing, updating, and validating employee records.

SOLUTION ARCHITECTURE DATA FLOW

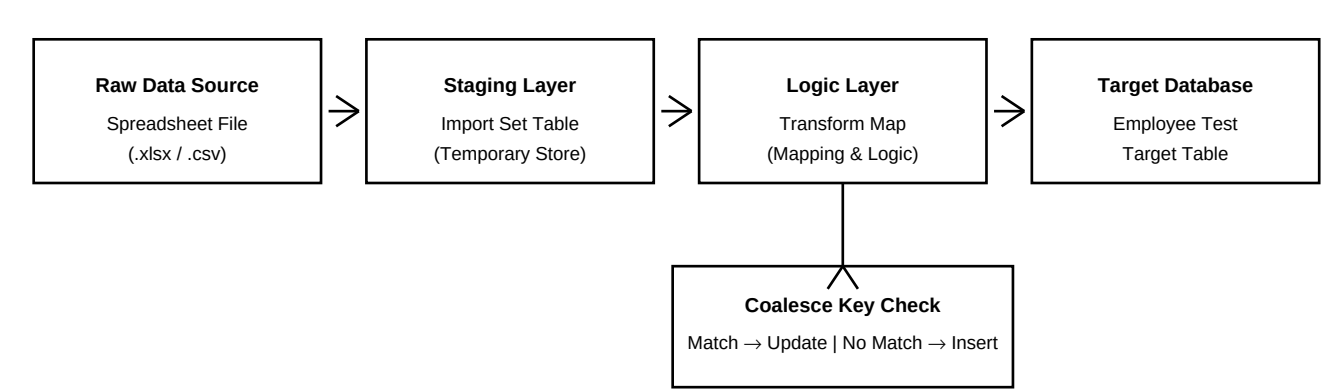


Figure 1: Architecture and data flow of the Import Data Using Transform Maps (Spreadsheet) process

Process Workflow & Technical Specification:

- 1. Data Source Setup:** The user or administrator uploads the external Excel or CSV file containing employee records.
- 2. Import Set Staging:** The spreadsheet data is loaded into the Employee Import Import Set Table, which acts as an intermediate staging area before transformation.
- 3. Transform Map Configuration:** Source fields are mapped to the Employee Test target fields using Auto-Map Matching or Mapping Assist. The configured Transform Map controls how the source values are transferred.
- 4. Coalesce Configuration:** Employee ID is used as the unique matching field. When a matching Employee ID exists, the existing record is updated; when no match exists, a new employee record is inserted.
- 5. Target Table Execution:** The transform process applies the configured mappings and commits the processed records into the Employee Test target table. Transform history provides visibility into inserted, updated, ignored, or unsuccessful rows.

References:

1. ServiceNow project documentation: Import Data using Transform Maps (Spreadsheet) — provided project details.
2. Project Design Phase – Module 3: Solution Architecture — provided reference document.
3. Project Design Phase – Solution Architecture — provided template.